



Project Title	Rising Waters: Machine Learning Based Flood Prediction System		
Developer	N. Joshnavi devi	Team ID	[Team ID]
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20. Scalability & Future Plan

The current system is deliberately scoped to what a solo developer can build and validate against a static, 115-record dataset within the project timeline. The items below represent the natural next steps once more data, more time, or additional infrastructure become available, and are grouped roughly from nearest-term to longer-term.

- Integrate live weather-API data feeds to replace manual/auto-filled input with real-time meteorological readings.
- Incorporate IoT/sensor-based river and water-level data for finer-grained, location-specific predictions.
- Add geospatial/map-based visualization of flood-risk zones across a region.
- Containerize the Flask application for cloud deployment and horizontal scaling.
- Introduce a retraining pipeline to periodically update the model as new historical data becomes available.
- Add user accounts and a historical-prediction dashboard for authorities to track trends over time.
- Extend delivery channels with SMS/email alerts for severe-risk classifications.

None of these require a rewrite of the existing architecture: the same train/serve separation documented in Section 10 means new data sources or a retraining cadence can be layered in without touching the request-handling code in app.py.

